

DUY TAN UNIVERSITY

GRADUATE SCHOOL



ASSIGNMENT

BLOCKCHAIN TECHNOLOGY



Supervisor : Dr. Anand Nayyar

Student: Trần Văn Bảo

Da Nang, May 2026

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Questions:

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Q1: What is Blockchain? What are its Various Types? Explain its working and what is the role of Miner? Explain some 4 Real Time use cases of Blockchain?

Answer:

Part 1: What is Blockchain?

To answer the first part, here is how I personally understand Blockchain in the simplest terms:

1. Real-Life Example: The “Drinking Fund”

Traditional way (Bank / Middleman): A group of friends gives all their money to one person who acts as the treasurer and keeps the record book. If that person becomes dishonest, changes the numbers, or accidentally loses the book, everyone loses their money.

Blockchain way: Nobody controls the record book alone. Instead, imagine there are 20 people in the group, and **everyone keeps their own identical copy** of the record book. Whenever someone spends money, they must announce it to the whole group, and all 20 people update their books with the exact same information at the same time.

2. Why is it so secure?

Now, imagine one person tries to cheat by changing their own record from *"I paid 500k"* to *"I paid 5 million"*.

When the group checks the records, the other 19 people will compare their books and immediately say: *"Your record is wrong. All of ours say you only paid 500k."* So, the fake change gets rejected instantly. To successfully hack the system, that person would need to secretly change the records of more than half of the group at the exact same time — which is almost impossible.

3. Simple One-Line Explanation

"Blockchain is basically a **shared digital record book**. Instead of one trusted company or bank controlling it, everyone in the network keeps the same copy. Once information is added, it cannot be easily changed, deleted, or faked."

Part 2: What are its Various Types?

For the second part, about the different types of Blockchain, I don't look at them as dry technical definitions. Instead, I group them based on **who is allowed to join and see the data**. From my understanding, there are 3 main types:

1. Public Blockchain (Open to Everyone)

What it is: This is a completely open network. Anyone in the world with an internet connection can join, look at the history of transactions, and participate without needing permission from anyone.

Real-life comparison: It is like a **public park**. Anyone can walk in, hang out, and see what is happening.

Examples: Bitcoin and Ethereum.

2. Private Blockchain (Restricted / Internal)

What it is: This is a closed network controlled by one specific organization. You cannot just join whenever you want; you must be invited by the owner. The data inside is private and only visible to authorized members.

Real-life comparison: It is like a **private company office**. Only employees with an ID card can enter and read company documents.

Why use it? Large companies use this to secure their internal data while still enjoying the fast and unchangeable benefits of Blockchain.

3. Consortium Blockchain (Group-Controlled)

What it is: This is a mix between public and private. It is not controlled by just one company, but it is also not open to the whole world. Instead, a small group of pre-selected organizations cooperate to run it.

Real-life comparison: It is like a **joint business project** between 4 or 5 different banks. They don't trust a single bank to hold all the records, so they share the control among themselves.

Part 3: How Blockchain Works & The Role of a Miner

1. How Blockchain Works (4 Simple Steps)

To understand how Blockchain works... let me explain it in a simple 4-step process when a transaction happens:

Step 1: let's say User A wants to send some money, or maybe some information, to User B. This request is broadcasted... I mean, sent to the whole network.

Step 2: Next, the system doesn't process just one transaction. It groups many transactions happening at the same time into a digital box. We call this box a "Block". It's like... writing on a new page of a notebook.

Step 3: Then, computers in the network—we call them Miners—they will check this transaction. For example, they check if User A really has enough money to send, to see if it's real.

Step 4: Finally, after checking, this block is locked and connected to the older blocks before it. This creates a very long chain, so that's why it is called "Blockchain". After this step, the transaction is finished and nobody can change or delete it anymore.

2. What is the Role of a Miner?

Now, about the Miner... because we don't have any bank or big company to control this network, we need Miners. In my opinion, they are like "digital accountants" of the system. They have two main jobs:

Job 1 is checking transactions: They use very powerful computers to check if transactions are real. This stops bad people from cheating the system.

Job 2 is solving math to lock the block: Blockchain has a rule, like... to add a new block to the chain, computers must race to solve a very hard math puzzle. The fastest computer wins the right to lock that block.

But why do they do this? They don't work for free, right? When a miner solves the puzzle and locks a block successfully, the system automatically rewards them with some new cryptocurrency, like free Bitcoin, and a small transaction fee from users. This whole process is what we call "Mining".

Part 4: 4 Real-Time Use Cases of Blockchain

Finally, about the real-time use cases... Blockchain is not just for Bitcoin, it actually has many practical uses in real life. Here are 4 examples that I understand:

1. Cryptocurrency (Sending Money)

This is the most famous use case, you know, like Bitcoin. Normally, if you want to send money to a friend in another country, you must go through a bank, and it takes days and high fees. But with Blockchain, you can send money directly to anyone in the world in just a few minutes. No banks needed, and it runs 24/7.

2. Supply Chain (Tracking Products)

Uhm, "supply chain" means tracking goods from the factory to the customer. For example, a supermarket can use Blockchain to track organic food from the farm. Because nobody can change the data on Blockchain, customers can just scan a QR code to see exactly where the food comes from and if it is real or fake.

3. Smart Contracts (Automatic Agreements)

I think this one is very cool. It's like... an automatic digital contract. For example, when you rent an apartment, you can use a smart contract. On the 1st of every month, the system will automatically take rent money from your account and send it to the landlord, and then it automatically gives you the digital door code. You don't need a real estate agent, so it saves money.

4. Voting Systems

And the last one is voting in elections. If we use Blockchain for voting, it will be 100% fair and honest. Why? Because once a citizen clicks "vote", that data is locked into the Blockchain forever. Nobody, not even the hackers or the government, can change or delete that vote. So, it stops cheating completely.

Q2: What is Bitcoin? Why bitcoin is the most traded Cryptocurrency? Explain the concept of Digital Wallet? And what are the major issues in Cryptocurrency?

Answer:

Part 1: What is Bitcoin?

Moving on to the second question, about "What is Bitcoin?"... here is my personal way to explain it simply:

Bitcoin is basically the very first cryptocurrency in the world. For me, I can think of it like... **digital cash** or even **digital gold**. It was created back in 2009 by a mysterious person or maybe a group of people named Satoshi Nakamoto. Today, nobody really knows who he is, which is quite interesting.

To make it easy to understand, it is like the real money we use every day, but it only exists on the internet. There are no physical coins or paper bills.

And because Bitcoin runs on the Blockchain technology that I mentioned earlier, it has one special thing: **it doesn't need a bank or a government to control it**.

Normally, if I want to send money to you, the bank must check and approve it, right? But with Bitcoin, I can send it directly from my digital wallet to your digital wallet, anywhere in the world, in just a few minutes. No middleman, no high fees, and nobody can freeze your account.

Part 2: Why Bitcoin is the Most Traded Cryptocurrency?

Now, for the second part... why is Bitcoin the most traded crypto? Like, why does everyone buy and sell it more than any other coin? In my opinion, there are 3 or 4 main reasons for this:

- **First, it has the "First-Mover" advantage.** Uhm, what I mean is Bitcoin was the very first cryptocurrency. It started in 2009, while other coins came much later. Because it has the longest history, people trust it the most. It is like the "King" or the "OG" of the crypto world.
- **Second, it has the biggest brand name.** Even people who don't know anything about technology, like my parents, they still know the word

"Bitcoin". So, when a new investor wants to enter the crypto market, Bitcoin is always their first choice.

- **Third, it is super easy to buy and sell.** Because it is the biggest coin, every crypto exchange in the world supports it. You can change Bitcoin into real money almost instantly, anywhere. In finance, we call this high liquidity... but basically, it just means it's very easy to trade.
- **And finally, big companies and banks are buying it.** Nowadays, Bitcoin is not just for tech people anymore. Big financial funds and institutions are holding a lot of Bitcoin, treating it like digital gold to protect their wealth.

So, yeah, because of trust, fame, and big money, Bitcoin remains the most traded cryptocurrency in the world.

Part 3: Explain the Concept of a Digital Wallet

For the third part, about the digital wallet... from my understanding, it is not like a real physical wallet. In crypto, there are no paper bills or physical coins inside your phone. All the coins always stay on the Blockchain network.

So, I see a digital wallet as a tool that holds two very important things: a **Public Key** and a **Private Key**. To make it easy to understand, I usually compare it to an email account:

- **First, the Public Key is like your email address.** You can give it to anyone so they can send you money. In the crypto world, we also call this your "wallet address."
- **Second, the Private Key is like your email password.** You use it to log in and approve transactions... like saying "Yes, I want to send this money." This key must be 100% secret because if you lose it, your money is gone forever. There is no bank to call, and there is no "Forgot Password" button here.

So, basically, I think a digital wallet is just a digital keychain. It doesn't hold the money, it only holds the keys to help you manage your money on the Blockchain.

Part 4: What are the Major Issues in Cryptocurrency?

And finally, about the major issues in cryptocurrency... Even though I think crypto is very smart, it still has some big problems in real life. In my opinion, there are 3 main issues right now:

- **The first issue is that the price changes too fast.** Uhm, in finance they call it "volatility," but basically, it means the price goes up and down crazily. For me, I actually lost a lot of money on Binance and BingX, even when I just held the coins. By the way, it was not about Bitcoin, but yeah... after that, I really learned a hard lesson. This extreme price change makes crypto very risky for normal users.
- **The second issue is security and scams.** Because crypto is all digital and has no central bank to control it, it is a dangerous place for hackers and scammers. If you lose your private key or accidentally send money to the wrong address, your money is gone forever. There is no customer support to help you fix the mistake.
- **And the last issue is about government rules.** Many countries still don't have clear laws for cryptocurrency. Some governments want to support it, but others want to ban it completely because they cannot control it. This makes the future of crypto very uncertain.

So, yeah, I think before cryptocurrency can become truly popular, it needs to solve these big problems about price stability, safety, and laws.

Q3: What is the concept of “Consensus”? Explain Proof of work and Proof of Stake and Enlist Differences between the two?

Answer:

Part 1: What is the Concept of “Consensus”?

Moving on to the third question, about the concept of "Consensus", to put it simply, consensus just means "agreement."

Like I mentioned before, Blockchain has no boss, no central bank, and no leader to decide what is right or wrong. So, how can thousands of computers around the world agree on the same history of money? They need a "Consensus Mechanism" which is basically a set of rules so everyone can agree on the same truth. If the majority of the network agrees a transaction is real, it becomes the truth.

Part 2: Explain Proof of Work and Proof of Stake

To achieve this agreement, there are two most popular ways:

- **Proof of Work (PoW):** This is the traditional way that Bitcoin uses. Like I explained in the first question, miners must use very powerful computers and a lot of electricity to race and solve hard math puzzles. The fastest computer wins the right to update the Blockchain.
- **Proof of Stake (PoS):** This is a newer way. From my understanding, PoS deletes the idea of "mining computers." Instead of buying expensive hardware, users must "stake"—which means locking up their own coins in the network. The more coins you lock up, the higher chance the system will choose you to verify transactions and get rewards.

Part 3: Differences Between the Two

To compare PoW and PoS, I see three major differences from my perspective:

- **First, about Resources:** PoW requires a lot of physical computer hardware and electricity. But PoS only requires money, meaning you just need to hold and lock your digital coins.

- **Second, about Environmental Impact:** Because PoW uses too much electricity, it is very bad for the environment. PoS is much greener and eco-friendly because it doesn't need heavy machines running 24/7.
- **Third, about Speed:** PoW is usually slow because solving math puzzles takes time. On the other hand, PoS doesn't have puzzles, so it is usually much faster and can handle more transactions.

So, that is how I understand Consensus and the two main ways it works.

Q4: What is Ethereum and explain System and Network Architecture in Brief? What is Hyperledger and write 10 Differences between Ethereum, Hyperledger, Bitcoin and Dogecoin?

Answer:

Part 1: What is Ethereum and its Architecture?

Now, moving to the fourth question, about Ethereum, from my understanding, if Bitcoin is just like a digital notebook for saving and sending money, then

Ethereum is like a giant global computer. It doesn't just hold money; it allows people to build apps and run smart contracts on it.

About its system and network architecture... I can explain it briefly like this:

- **First, it uses a Peer-to-Peer (P2P) Network:** This means thousands of computers around the world we call them "nodes" connect directly to each other. There is no central server.
- **Second, it has the EVM (Ethereum Virtual Machine):** You can think of the EVM as the "**main brain**" or the **engine** of the Ethereum computer. When someone writes a smart contract, the EVM is the one that reads and runs that code on every computer in the network to make sure the result is the same.
- **Third, the Gas system:** Like to make a real computer work, you need electricity. On Ethereum, to run code or make transactions, you must pay a small fee called "Gas" using its coin, Ether (ETH). This stops people from spamming the system with bad code.

Part 2: What is Hyperledger?

Next, about Hyperledger. This one is completely different from Ethereum or Bitcoin.

In my opinion, Hyperledger is **not a cryptocurrency**. It has no coin, and you cannot buy it on Binance. It is actually an open-source project created by the Linux Foundation to help big corporations build their own **private blockchains**.

For example, a group of banks or supply chain companies can use Hyperledger to share private data safely with each other, without showing it to the public world.

Part 3: 10 Differences between Ethereum, Hyperledger, Bitcoin, and Dogecoin

To compare these four, I don't want to use boring technical tables. Instead, here are 10 clear differences based on how I understand them:

1. **Primary Purpose:** Bitcoin is for digital money; Ethereum is for running apps; Hyperledger is for business privacy; and Dogecoin... well, it started just as an internet joke or a meme.
2. **Blockchain Type:** Bitcoin, Ethereum, and Dogecoin are **Public** (anyone can join). Hyperledger is **Private/Permissioned** (only invited businesses can join).
3. **Native Currency:** Bitcoin has BTC; Ethereum has ETH; Dogecoin has DOGE; but Hyperledger has **no native coin** at all.
4. **The Creator:** Bitcoin was made by Satoshi (we don't know him); Ethereum by Vitalik Buterin; Hyperledger by the Linux Foundation; and Dogecoin by two software engineers as a joke.
5. **Smart Contracts:** Ethereum and Hyperledger are built for complex smart contracts. Bitcoin and Dogecoin do not really support them—they are just for sending money.
6. **Transaction Speed:** Bitcoin is very slow; Dogecoin and Ethereum are faster; but Hyperledger is **the fastest** because it is private and doesn't need global mining.
7. **Consensus Mechanism:** Bitcoin and Dogecoin still use Proof of Work (heavy mining). Ethereum uses Proof of Stake (staking coins). Hyperledger doesn't use coins, it uses business agreements (like PBFT).
8. **Coin Supply:** Bitcoin is strictly limited to 21 million coins. Dogecoin has an **infinite supply** (new coins are made forever). Ethereum is flexible, and Hyperledger has zero.

9. **Transaction Fees:** On Bitcoin, Ethereum, and Dogecoin, you must pay fees to miners or validators. On Hyperledger, because it's a private business network, there are no traditional crypto transaction fees.
10. **Target Target Users:** Bitcoin and Dogecoin are for regular investors and internet users. Ethereum is for software developers. Hyperledger is strictly for big enterprises and corporations.

Q5: Elaborate NFT Tokens? Explain DEFI and also illustrate how Blockchain can support Cyber Security in Real time world?

Answer:

Part 1: Elaborate NFT Tokens

To answer the fifth question, let's start with NFTs. For me, I think NFT stands for Non-Fungible Token, but honestly, that name sounds too academic. To make it simple, I see an NFT as a **digital birth certificate** or a digital receipt for a unique item.

Many people confuse Bitcoin and NFTs, but from my understanding, they are very different:

- Bitcoin is "fungible" which means interchangeable. If I borrow a 100-dollar bill from you, I can return a different 100-dollar bill, and it's totally fine because they have the same value.
- But an NFT is "non-fungible" meaning it is 100% unique. It is like a football signed by Messi, or a real estate certificate for a specific house. You cannot just swap it for another one.

So, when someone buys an NFT, like a digital artwork or a rare item in a video game, the Blockchain proves that they are the **only true owner** of that specific digital asset. Nobody can copy or fake that ownership.

Part 2: Explain DEFI (Decentralized Finance)

Next, about DeFi, this means Decentralized Finance. Basically, I think of it as a financial system or a bank, but **without the bank building and without the bankers**.

Normally, if you want to borrow money, get a loan, or save money to earn interest, you must go to a traditional bank. You have to fill out many papers, wait for approval, and pay fees to the middleman, right?

But with DeFi, everything is controlled automatically by smart contracts on the Blockchain. For example, if I want to borrow money, I can just go to a DeFi

application, lock some of my crypto as collateral, and the system instantly gives me a loan in a few seconds. There are no credit checks, no human bias, no paperwork, and it runs 24/7. It makes financial services open to everyone in the world.

Part 3: How Blockchain Supports Cyber Security

And finally, how Blockchain can support Cyber Security in the real-time world. I think this is a huge deal because traditional security is failing a lot nowadays.

Right now, most companies save all their important data in one central server. If a hacker breaks into that server, everything is stolen or deleted, right? But Blockchain changes this game in a few ways:

- **First, no more single point of attack:** Because Blockchain is decentralized, data is shared across thousands of computers. A hacker cannot just attack one company website to steal everything; they would have to hack more than half of the network at the same time, which is almost impossible.
- **Second, hackers cannot hide their tracks:** In traditional systems, when a hacker breaks in, they often change the history logs to delete evidence. But on Blockchain, once data is recorded, it is locked forever. So, security teams can track unauthorized changes instantly in real time.
- **Third, safer logins:** Instead of using traditional passwords that can get leaked from a company's database, we can use the Public and Private Key system of Blockchain to log into accounts securely. No passwords saved on servers means hackers have nothing to leak.

So, by removing the central database, I believe Blockchain makes the internet much safer from cyber attacks.